

Task 2 — Test Plan: Product Recommendation Carousel (Next.js)

A list of test cases a team would implement for a **product carousel** whose data is served to the frontend by a **recommendation platform's API**, focused on the tests that run **during the build / CI process** of the Next.js application.

This document only **lists** the test cases, their **type**, and their **pass criteria** — it does not implement them.

Guiding principle

Tests that run in the build/CI pipeline must be **deterministic**, so they **do not call the live recommendation API** (external, flaky, rate-limited, and non-reproducible). Instead:

- **Build-time tests** run against **recorded fixtures / mocks** — they verify *"does our code behave correctly given a known response?"*

Unless stated otherwise, every test below assumes a **mocked API response**.

1. Static analysis / type safety

#	Test case	Type	Pass criteria
1.1	Project type-checks	Static (tsc --noEmit)	Exits 0; no type errors
1.2	Lint & format	Static (ESLint / Prettier)	0 errors; no violations
1.3	API response is strongly typed	Static	No any leaks from the fetch/data layer into components

2. Data mapping (API to view model)

#	Test case	Type	Pass criteria
2.1	Price formatting	Unit	Raw price → A\$69.95 (correct currency, locale, 2 decimals)
2.2	Discount label derivation	Unit	Discount data → "GET \$10 OFF" (or absent when no discount)

#	Test case	Type	Pass criteria
2.3	Swatch overflow ("+N more")	Unit	6 colours → renders 3 swatches + "+3 more"
2.4	Missing optional fields	Unit	No swatches / no discount → maps to safe defaults, no throw
2.5	Product → card model shape	Unit	Mapper output equals expected view model for each fixture

3. Contract / schema validation (against fixtures)

#	Test case	Type	Pass criteria
3.1	Valid response parses	Schema (Zod / JSON-Schema)	Recorded fixture satisfies the schema
3.2	Required fields present	Schema	id, name, price, image required; absence fails the parse
3.3	Malformed / partial response	Schema	Rejected or coerced to a safe fallback — never crashes render
3.4	Unexpected extra fields	Schema	Ignored gracefully; does not break parsing

4. Component rendering & interaction

#	Test case	Type	Pass criteria
4.1	Renders N cards	Component (RTL)	Exactly N cards for N items in the response
4.2	Card content	Component	Each card shows image, name, price, discount badge, wishlist icon,

#	Test case	Type	Pass criteria
			swatches
4.3	Empty recommendations	Component	Carousel not rendered (or fallback shown); no crash
4.4	Single item	Component	Nav arrows hidden/disabled; no broken looping
4.5	Next / Prev navigation	Component	Advances / regresses the visible window correctly
4.6	Start boundary	Component	On first slide, Prev is disabled or loops (per design spec)
4.7	End boundary	Component	On last slide, Next is disabled or loops (per design spec)
4.8	Favrouite toggle	Component	Click fires handler / toggles pressed state
4.9	Lazy image loading	Component	Offscreen images are not eagerly loaded

5. Async states (loading / error / empty)

#	Test case	Type	Pass criteria
5.1	Loading state	Component	Skeleton/placeholde r shown while the request is pending
5.2	Error state	Component	API failure → graceful fallback via error boundary; no white-screen
5.3	Empty state	Component	Empty array → carousel hidden or fallback message shown

6. Accessibility

#	Test case	Type	Pass criteria
6.1	No automated a11y violations	a11y (jest-axe)	0 axe violations
6.2	Carousel semantics	a11y	role="region" + aria-roledescription="carousel"; slides labelled
6.3	Control labels	a11y	Prev/Next and wishlist buttons have accessible names
6.4	Keyboard operability	a11y	Controls reachable/operable by keyboard; focus order logical; focus visible
6.5	Image alt text	a11y	Product images have meaningful alt; decorative icons aria-hidden
6.6	Reduced motion	a11y	prefers-reduced-motion disables auto-advance / animated transitions

7. Responsive / layout

#	Test case	Type	Pass criteria
7.1	Cards per breakpoint	Component (parameterised)	Visible card count matches spec per breakpoint (e.g. 1–2 mobile → 4 desktop)
7.2	No horizontal overflow	Component / visual	Page body never scrolls horizontally
7.3	Touch / swipe	Component	Swipe advances slides on touch viewports

8. Next.js-specific

#	Test case	Type	Pass criteria
8.1	next/image sizing	Integration	Images use width/height (or fill) to reserve space → no layout shift
8.2	Image host config	Integration/build	Remote image host is whitelisted in next.config images
8.3	No hydration mismatch	Integration	Server and client markup match for the carousel (no hydration warnings)
8.4	SSR/ISR fetch failure	Integration	A data-fetch failure degrades to a fallback rather than failing the whole page
8.5	Production build succeeds	Build smoke test	next build completes and exits 0